Question and Answer
Mobile Internet Technology

Study Guide Reference

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Contents

1 Physical Layer **2**

2 MAC Layer **5**

3 Wireless Data Networks: WLAN/Wi-Fi **7**

4 Multi-hop Networks **8**

5 Mobile Networks: xG **10**

6 Internet Protocols & Mobility **12**

1 Physical Layer

Q: What are the challenges of mobile com.?

A: Restricted freq. band, low transmission rates, higher loss rate/latency/Jitter, lower security.

Q: How does the Freq. impact the antanna size?

A: Smaller Freq. = Larger antannas.

Q: Which signals do you mix for analog modulation?

A: Message and carrier signal.

Q: How does PSK work?

A: Phase of the carrier signal is changed to represent data.

Q: What is one big disadvantage in phase shit keying?

A: Noise has a high impact on the signal.

Q: Is a higher or lower SNR better?

A: Higher.

Q: What is the formula to calculate the signal propagation/recieved signal power?

A: P_r proportional to $1/d^2$.

Q: What can the received power get influenced by?

A: Distance, fading, shadowing, antenna gain.

Q: Assume that you have given binary PSK (BPSK) and QPSK for doing digital modulation. Which advantage(s) has BPSK compared to QPSK?

A: Using BPSK is more robust against transmission errors than using QPSK.

Q: Why does a given bandwidth limit the data rate one can achieve on a wireless channel?

A: Bandwidth restricts the signal rate.

Q: Why is the data rate you can achieve in a wireless system depending on the distance between sender and receiver?

A: Receive power decreases with distance, The signal-to-noise ratio typically is smaller for larger distances, More robust modulation has to be used to encode the data.

Q: You are asked to set up a communication system which operates on a bandwidth of 20 MHz, and you have given a signal-to-noise ratio of 30 dB. What is approximately the maximum data rate you can achieve with this system?

A: 200 MBit/s.

Q: Assume that in a given wireless system a sender and a receiver are placed in a certain distance to each other. Which actions might help to increase the received signal power?

A: Use a lower carrier frequency, Increase transmit power, Increase antenna gain of sender and receiver.

Q: How does the receive power P_r in vacuum depend on the distance d between sender and receiver?

A: $P_r \sim 1/d^2$.

Q: Which techniques/effects are involved in causing inter-symbol interference?

A: Multipath propagation, Time dispersion.

Q: How can you protect against inter-symbol interference?

A: Reduce signal rate, Use of guard times between signals.

Q: How does OFDM increase the robustness of the transmission?

A: OFDM divides the bandwidth b into n sub-bands on which several narrowband signals are transmitted simultaneously.

Q: What is space division multiplexing and how is it implemented using cell structures?

A: Space division multiplexing uses cell structures, where each base station covers a specific transmission area (cell). Mobile stations communicate only via their base station.

Q: What are the main advantages and problems of using cell structures?

A: Advantages: Frequency re-use, higher capacity, less transmission power needed. Problems: Requires network infrastructure, handovers are necessary, interference between adjacent cells.

Q: How to calculate the packet error probability?

A: $P_P = 1 - (1 - BER)^n$ for frames of n bit length.

Q: What is the code rate when using shift registers?

A: k/n where k is input bits and n is output bits. 1/3 means 3 bits are produced for every 1 input bit.

Q: How can you mitigate the effects of narrowband interference?

A: Use of signal spreading.

Q: Which technique is more robust against inter-symbol interference?

A: Orthogonal Frequency Division Multiplexing (OFDM).

Q: In the lecture, some dimensions have been defined, in which multiplexing is done. These dimensions are:

A: Time, Frequency, Code, Space.

Q: When using Time Division Multiple Access (TDMA) in large cells, a so-called timing advance can help improving the performance of the network. Why is this done?

A: All stations get a common understanding of the used time frame.

Q: You have given a CDMA system working with quasi-orthogonal codes, and you use BPSK for modulation. How can you reduce the error rate for transmissions of one specific device?

A: Increase transmit power of this device, Use longer codes for transmissions of this device.

Q: Why have simple convolutional codes today been replaced by turbo codes?

A: The decoding using convolutional codes is very slow when errors have been present.

Q: What does a code rate of 1/3 mean?

A: Two thirds of the transmitted bits is overhead caused by error correction.

Q: Comparing PSK and ASK for doing digital modulation, which one is preferable for data transmission? Give reasons.

A: PSK, because it is more robust against attenuation and noise which affect the amplitude.

Q: In reality, modulation is done in two steps: first a digital modulation followed by an analog modulation step. What is the purpose of each of these steps?

A: Digital modulation converts bits into an analog baseband representation within a restricted bandwidth. Analog modulation upconverts this baseband signal to a specific high carrier frequency/channel.

Q: What is the near/far problem in CDMA, and how is it solved?

A: Near devices overpower far devices at the base station. Solved using precise, high-frequency dynamic power control to ensure all signals arrive with equal strength.

Q: What problem does Timing Advance solve in large TDMA cells, and how?

A: Solves propagation delay where distant signals arrive late and overlap into other time slots. The BS measures delay and instructs the device to transmit earlier.

2 MAC Layer

Q: What is the problem with idle listening in mobile internet?

A: It wastes huge amounts of energy.

Q: What does CSMA/CD stand for? Also explain it

A: Carrier Sense Multiple Access/Collision Detection.

Q: Name some energy problems (i.e. what wastes energy?)

A: Collisions, Overhearing (listening to a packet destined for another node), idle listening, protocol overhead.

Q: Define Centralized Medium Access and in (dis-)advantages

A: A central station defined when a node may access the medium. Advantages: Simple and efficient (no collisions). Disadvantages: Burdens the central station, signaling overhead and delays.

Q: Which distributed medium access options exist? Describe them

A: Schedule-based (via TDMA, no collisions/idle listening issues but sync needed) and Contention-based (accept risk of collisions to avoid coordination overhead).

Q: What is a busy tone?

A: A signal that is used to check if the medium is in use.

Q: Which of the following medium access control approaches in theory have no hidden station problem when used in a wireless network?

A: Reservation Aloha, Wireless Collision Detect, MACA.

Q: On which reason slots are used in Slotted ALOHA?

A: Reduction of the effects of collisions.

Q: Why can MACA fail solving the hidden station problem?

A: Channels can be asymmetric, Moving stations are present.

Q: What's the conceptual difference between busy tones and RTS/CTS for collision avoidance?

A: Busy tones use out-of-band passive occupancy detection on a separate frequency, while RTS/CTS is an active, in-band reservation protocol using control packets.

3 Wireless Data Networks: WLAN/Wi-Fi

Q: What is the difference between infrastructure network and ad-hoc networks?

A: In infrastructure networks, devices communicate via a central device like a router or access point. In ad-hoc networks, devices communicate directly with each other without any central infrastructure.

Q: Explain mesh networks

A: Similar to infrastructure networks but wireless APs/Mesh routers are allowed to forward data to each other.

Q: How does WiFi MAC handle transmission coordination?

A: Uses CSMA/CA to manage hidden/exposed stations, offering both best-effort asynchronous service and polled time-bound QoS.

Q: How does prioritized media access control work in 802.11 WLANs using SIFS, PIFS and DIFS?

A: We have three priorities (SIFS/PIFS/DIFS) that define how fast we can send after the medium becomes idle. SIFS is for high-priority control like ACKs, DIFS for normal data.

Q: Regarding the CSMA/CA Method, what happens if we draw a random backoff time but dont get medium access?

A: In the next interval the REMAINING backoff time is used.

Q: What is frame aggregation?

A: Assemble several frames of one station and remove redundant header information.

Q: For which purpose, DSSS is used in 802.11?

A: Increased robustness of transmission.

Q: What is the standard medium access variant in 802.11?

A: CSMA/CA.

Q: By which means the data rate in newer Wi-Fi variants is increased compared to older variants (802.11a/b/g)?

A: Larger bandwidth, MIMO, Increased FEC coding rate, Modulation like 256-QAM or higher.

Q: Why are the different waiting times DIFS, PIFS and SIFS used in medium access of 802.11?

A: To prioritize control messages like ACKs over data transmission.

Q: To safe energy, stations can switch off (sleep) if they have nothing to transmit. In infrastructure mode, who is buffering data for stations which are currently sleeping?

A: The Access Point.

Q: The PLCP header in 802.11 contains address fields. Why are multiple address fields necessary?

A: To allow for not only including sender and receiver address, but also an identification of the AP which forwards the frames within a distribution system.

4 Multi-hop Networks

Q: What is the key difference between a MANET and a WMN?

A: MANET (Mobile Ad-hoc Network): Each mobile device acts as a mobile router. WMN (Wireless Mesh Network): Has a backbone of usually static wireless routers.

Q: What are the traditional routing algorithms? Describe them also

A: Distance Vector: Routers exchange info only with neighbors, slow convergence, risk of routing loops. Link State: Routers exchange complete topology info via flooding, fast but high overhead.

Q: What are some problems of traditional routing algorithms

A: We have Frequent changes of available links, connection quality and maybe changes of participating nodes, as well as asymmetric connections.

Q: Name req. for Routing Protocols

A: Self-starting, self-organizing, dynamic topology, minimal traffic overhead, rapid route convergence, low resource consumption, scalable.

Q: Why is hop count often not a suitable metric

A: Asymmetric links, data loss ratio, and varying data rates on different links.

Q: What is ETX? Describe it

A: Expected Transmission Count: Expected number of transmissions required for successfully transmitting a packet over a link. ETX = 1 / (FDR * RDR).

Q: What is ETT? Explain it

A: Expected Transmission Time: ETT = ETX * (S / B) where S is average packet size and B is bandwidth.

Q: What is Weighted Cumulative ETT? Explain it

A: Enhancement of ETT (WCETT) that handles channel diversity and interference in multi-radio mesh networks.

Q: Describe Reactive, Proactive and Hybrid topology

A: Reactive: Discover routes on demand. Proactive: Maintain all routes continuously. Hybrid: Combine both.

Q: What are the differences between proactive and reactive topology?

A: Proactive has low latency for route discovery but high overhead. Reactive has high initial latency but low overhead if traffic is low.

Q: What are the ways to optimizations Dynamic Source Routing (DSR)?

A: Topology Learning, Controlled Flooding, Route Caching, and Reply Storm Prevention.

Q: Name the (dis-)advantages of DSR

A: Advantages: No periodic updates, fast if cached, redundant paths. Disadvantages: Initial delay, bloated headers due to full path storage.

Q: How does AODV work?

A: AODV (Ad-hoc On-Demand Distance Vector): Nodes store only the next-hop info in local routing tables. Uses sequence numbers to prevent loops.

Q: Describe OLSR

A: Optimized Link State Routing: Uses Multipoint Relays (MPRs) which are selected 1-hop neighbors that cover all 2-hop neighbors to reduce flooding overhead.

Q: How does DSDV improve upon traditional Distance Vector routing for ad-hoc networks?

A: Introduces sequence numbers to prevent routing loops and handle dynamic topologies in an orderly manner.

Q: Why does AODV use sequence numbers?

A: To prevent routing loops and ensure freshness of routes.

Q: Which of the following routing protocols support assymetric links?

A: DSR supports asymmetric links via reverse route discovery.

Q: What is the key conceptual difference between AODV and DSR?

A: AODV uses table-driven next-hop routing (stateful), while DSR uses source routing where the entire path is carried in the packet headers (stateless).

5 Mobile Networks: xG

Q: Explain the cell concept

A: Hexagonal cell pattern is idealized. Cell sizes depend on the environment and traffic density. Allows frequency re-use across distant cells.

Q: What is the difference between the home location register and visitor location register?

A: HLR: Central database storing all master subscriber data and current location area. VLR: Local database linked to an MSC storing data of all users currently roaming in that area.

Q: What is HSCSD ?

A: HSCSD (High-Speed Circuit Switched Data): Combines multiple GSM time slots (up to 4) to achieve higher data rates, billed by time.

Q: Why is it better to use Quasi Orthogonal on the uplink and Orthogonal on the downlink?

A: Synchronization is required for orthogonal codes, which is simple on the downlink (one sender), but complex on the uplink where multiple independent devices transmit.

Q: What is the problem with cell breathing?

A: As cell traffic/noise increases, the effective coverage radius of the cell shrinks, causing distant users to drop out.

Q: What is macro diversity in UMTS and how does it work?

A: Simultaneous connection of a mobile terminal to up to 3 Node Bs to improve signal quality during soft handovers.

Q: What were the main improvements in LTE?

A: Higher data rates, elimination of CDMA in favor of OFDMA, Voice over LTE (VoLTE), and a pure packet-switched IP core network.

Q: Erkläre HSCSD und GPRS

A: HSCSD represents circuit-switched data with stable fixed channels billed by time. GPRS is packet-switched data with dynamic slots on demand billed by volume.

Q: What is the purpose of Paging?

A: Locating a mobile station within a location area via a broadcast call before establishing a connection.

Q: What is a network slice?

A: A virtual network instance running on top of a single physical network infrastructure simultaneously with other virtual network instances.

6 Internet Protocols & Mobility

Q: What is Complementary Code Keying (CCK) in wireless communications, and how does it differ from Barker code/DSSS?

A: CCK uses multiple codes to encode information (unlike Barker code which uses one fixed code). It uses 8-chip spreading sequences and enables higher data rates.

Q: What is the problem of TCP/IP stack regarding mobility

A: The IP address serves as both locator and identity simultaneously. Transport-layer connections (TCP) break whenever the IP address changes during roaming.

Q: What are the Req. for mobile ip?

A: Transparency (keep IP, continue session), Compatibility (no changes to normal routers), Security, and Efficiency.

Q: Explain mobile IP and the components added for it

A: MN retains its Home IP. When roaming, it receives a Care-of Address (COA). Home Agent (HA) intercepts and tunnels packets to the COA. Foreign Agent (FA) delivers them.

Q: What are the three encapsulation methods in Mobile IP and their key characteristics?

A: 1. IP-in-IP (Mandatory, 20B overhead), 2. Minimal Encapsulation (Optional, 8-12B overhead, no fragmentation), 3. GRE (Generic Routing Encapsulation).

Q: What is triangular routing problem and how to solve it?

A: Packets to the MN must go through the HA, while packets from the MN go directly to the CN, causing latency. Solved via Route Optimization where CN learns the COA.

Q: How does IPv6 solve/simplify mobility support?

A: Mobility is natively supported via extension headers, elimination of separate FAs (uses standard router advertisements), and automatic route optimization[cite: 1].

Q: What is the Host Identity Protocol (HIP) and its key features?

A: Decouples identity from location by introducing a cryptographic Host Identity (HI) layer between the network and transport layers[cite: 1].

Q: Explain credit-based authentication

A: Limits the volume of data a responder sends to an unverified new locator address until it receives acknowledgment, preventing reflection attacks[cite: 1].

Q: What makes QUIC faster than TCP+TLS?

A: 0-RTT handshake on subsequent connections, UDP-based deployment (no kernel dependency), and built-in TLS 1.3 encryption[cite: 1].

Q: How does QUIC handle network changes?

A: Uses 64-bit Connection IDs (CIDs) instead of IP/port bindings. When the network changes, the CID remains the same, keeping the connection alive seamlessly[cite: 1].